Paper revisions for supervisor review

Replacement passages and verified recall results

9 September 2026

Scope. This pack provides replacement passages for the supplied manuscript. The primary cultural result and original model-level recall intervals are preserved. Newly calculated comparisons are labelled exploratory. The original PDF has not been altered.

Main change. Explain matching decisions before recall, then present RQ1 (WEIRD–Serbian), RQ2 (model performance), RQ3 (US–China), and descriptive judge checks in that order. The accompanying `revision_notes.md` explains every change and gives reproducibility details. `replacement_sections.tex` contains the editable passages.

Category correction. The table separates five Frontier/API systems, six larger open-weight systems, and two smaller open-weight systems. Qwen 3.7 Max moves from the original mixed middle tier to the API group: official documentation describes a commercial API flagship, and a public-weight release was not verified. The categories describe access and relative size, not a capability ranking. Original-tier summaries remain in the analysis files for comparison; no model is dropped.

Integration. Copy the indicated passages into the current source, update cross-references and the model-category definition, and move the original precision/F1 table into the supplementary material. The unaffected introduction, related work, topic construction, prompts, bibliography and other original material remain in the submission source.

Abstract: replacement result sentences

Mean recall was 0.581 for WEIRD topics and 0.594 for Serbian topics. Across the eight matched topic pairs, the mean WEIRD-minus-Serbian difference was −0.013, or −1.3 percentage points (95% bootstrap CI: −8.1 to 5.8 percentage points). An exact one-sided Wilcoxon signed-rank test on the eight pair-level differences provided no evidence for the hypothesised WEIRD recall advantage ($p = 0.680$).

Introduction: replacement research questions

We address three research questions:

RQ1 (primary). Do LLMs recover fewer documented perspectives on Serbian sociopolitical topics than on structurally matched WEIRD topics? We hypothesise higher recall on the WEIRD topics.

RQ2 (exploratory). How does perspective recall vary across individual models, frontier API systems, and larger and smaller open-weight systems?

RQ3 (exploratory). How do the selected models from US and Chinese developers differ in overall recall and in their WEIRD–Serbian recall gaps?

Recall is the principal outcome for all three questions because the study concerns recovery of a fixed set of documented perspectives. Precision and F1 are supplementary descriptions of matching behaviour. The additional category and developer-country comparisons are exploratory.

3 Methodology: replacement subsections

3.3 Judge evaluation

The judges make matching decisions; recall is calculated after these decisions have been collected. For each model and topic, the model produces five perspectives. Each of five LLM judges receives one generated perspective at a time, together with the topic's five reference perspectives. The judge selects the reference that expresses the same underlying stance, or returns *no match*. This produces five decisions per generated perspective and 25 decisions per model–topic response when all calls succeed. Judges do not receive one another's decisions.

We use *family-disjoint judging*: no judge belongs to the same developer family as the generator whose output it evaluates. Here, a family is the developer grouping used in the evaluation configuration. For example, Gemini and Gemma belong to the Google family, GPT and GPT-OSS to OpenAI, and Claude Opus and Claude Sonnet to Anthropic. The five judges in each realised panel also come from distinct developer families.

The base panel consists of Gemini 3.1 Pro (Google), GPT 5.5 (OpenAI), Claude Sonnet 5 (Anthropic), DeepSeek V4 Pro (DeepSeek), and Kimi K2.6 (Moonshot). If the generator shares a family with a base-panel judge, we replace that judge with a designated substitute. Grok 4.5 replaces the Google, OpenAI, Anthropic, or Moonshot judge when necessary; Qwen 3.7 Max replaces the DeepSeek judge. Other generators use the unchanged base panel. This procedure prevents same-family evaluation, but does not establish that the judges are free from shared biases.

The reference order is randomised independently for every judgment. Returned indices are mapped back to the original reference identifiers before scores are calculated. Of 5,200 intended decisions, 5,184 are valid after retries. There are 1,024 generated perspectives with five valid judgments and 16 with four. A failed call is treated as missing data; a valid no-match decision is retained and contributes zero credit.

3.4 Calculating perspective recall

Recall measures the proportion of the five reference perspectives recovered in a model's five outputs. We use it as the primary metric because the evaluation target is a finite set of documented positions, and RQ1 concerns which of those positions are omitted. The five-output budget gives every model the same opportunity to recover that set. Repeating a position uses an output slot without covering an additional reference.

Step 1: interpret one judge's assignments. Suppose a judge assigns the five generated perspectives to

$$1, \quad 2, \quad 2, \quad 4, \quad \text{no match.}$$

Three distinct references have been recovered: 1, 2, and 4. This judge's recall for the model–topic response is therefore

$$R_j = \frac{\text{number of distinct references recovered}}{5} = \frac{3}{5} = 0.60.$$

The repeated match to reference 2 does not increase recall. Although four outputs match a reference, they cover only three distinct positions.

Step 2: combine the judges' assessments. When all five judges evaluate all five outputs, topic recall equals the mean of their five distinct-reference recalls:

$$R_{mt} = \frac{1}{5} \sum_{j=1}^{5} R_{mtj}.$$

For example, judge recalls of 0.60, 0.60, 0.80, 0.40, and 0.40 give topic recall 0.56. This corresponds to an average of 2.8 of the five references recovered across judges. It does not require the judges to agree on a single set of recovered references, and it does not apply a majority-vote threshold.

Exact aggregation, including missing judgments. The implementation combines fractional reference credits and accounts for missing judgments separately for each output. For a fixed model and topic, let y_{jk} be judge j's assignment for generated perspective k, where references are indexed by $g \in \{1, \ldots, 5\}$ and zero denotes no match. Let n_{jg} be the number of available assignments by judge j to reference g. Each assignment receives credit

$$c_{jkg} = \begin{cases} 1/n_{jg}, & y_{jk} = g, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Thus, two matches to the same reference receive one half each; together they contribute one unit for that judge. If J_k contains the judges with a valid decision for output k, the panel credit is

$$w_{kg} = \frac{1}{|J_k|} \sum_{j \in J_k} c_{jkg}. \tag{2}$$

We sum these credits for each reference, cap the sum at one, and average across the five references:

$$C_g = \min\left(1, \sum_{k=1}^{5} w_{kg}\right), \qquad R_{mt} = \frac{1}{5} \sum_{g=1}^{5} C_g. \tag{3}$$

With complete panels, these equations reduce to the mean judge recall given above. When judgments are missing, $|J_k|$ can differ across outputs; the cap prevents any reference receiving more than full credit. All outputs in this dataset have at least four valid judgments.

Step 3: summarise topics and models. Each topic receives equal weight. Overall model recall is the mean of the 16 topic recalls. WEIRD recall and Serbian recall are the corresponding means over eight topics each. Therefore,

$$\overline{R}_m = \frac{1}{16}\sum_{t=1}^{16} R_{mt} = \frac{\overline{R}_m^{\mathrm{W}} + \overline{R}_m^{\mathrm{S}}}{2}. \tag{4}$$

Recall describes coverage of the benchmark's reference set. It does not measure coverage of every legitimate position that may exist outside it.

3.5 Statistical analysis

RQ1: the primary matched-topic comparison. For each generator $m \in \{1, \ldots, 13\}$ and matched topic pair $p \in \{1, \ldots, 8\}$, we calculate

$$d_{mp} = R_{mp}^{\mathrm{W}} - R_{mp}^{\mathrm{S}}. \tag{5}$$

Positive differences favour WEIRD topics. The resulting 13×8 matrix contains 104 paired differences. We average across generators within each pair,

$$\Delta_p = \frac{1}{13}\sum_{m=1}^{13} d_{mp}, \tag{6}$$

and apply an exact one-sided Wilcoxon signed-rank test to the eight Δ_p values. The directional alternative is a positive shift, corresponding to the predicted WEIRD advantage. We also report the two-sided test. The signed-rank inference assumes independent pair-level differences with a symmetric distribution under the null; it is not a direct test of the arithmetic mean. The mean difference and its 95% percentile bootstrap interval summarise effect size. The interval uses 10,000 resamples of the eight matched pairs, keeping all 13 models together within each sampled pair.

RQ1: exploratory sensitivity analyses. We examine the alternative aggregation across pairs within each model, $\overline{d}_m = \frac{1}{8}\sum_p d_{mp}$, and apply an exact signed-rank calculation to these 13 model means. We remove numerical roundoff before ranking, use average ranks for ties, and enumerate every sign assignment. This comparison is conditional on the same eight topic pairs and does not make the selected model families independent cultural replications.

To examine variation across both dimensions, we also resample generator rows and matched-pair columns independently with replacement, retaining the Cartesian-product submatrix in each of 10,000 replicates. This crossed bootstrap provides an approximate interval for the mean gap.[1] A supplementary crossed random-intercept model provides a separate model-based check:

$$d_{mp} = \mu + u_m + v_p + \epsilon_{mp}, \tag{7}$$

with independent random intercepts for generator and matched pair and an independent residual. We estimate it by restricted maximum likelihood and report an approximate two-sided Wald test for μ. These analyses account for the two grouping dimensions under their respective assumptions; they do not establish representativeness of the selected models and topics or explicitly model dependence between different generators from one family. We do not treat the 104 differences as independent observations.

RQ2: model performance. The main table reports overall recall and recall separately for WEIRD and Serbian topics. We distinguish five Frontier/API systems without a verified public-weight release for the evaluated model, six larger open-weight systems, and two smaller open-weight systems. Qwen 3.7 Max is included in the API group, following its documentation as a commercial API flagship,[2] rather than in the original mixed Strong/open-weight tier. The smaller open-weight

[1] The resampling scheme follows Owen (2007), *The pigeonhole bootstrap*, arXiv:0712.1111.

[2] See Alibaba Cloud's Qwen3.7-Max commercial-model announcement. Public-weight release sources for the eight open-weight generators are listed in the accompanying revision notes.

group contains Gemma 4 31B and Llama 3.3 70B; the remaining six public-weight systems form the larger open-weight group. These are operational access and relative-size categories, not a capability ranking. All generators were evaluated through the same API-based workflow, whether or not their weights are public. Weight availability describes the model release, not how its outputs were generated in this experiment. Models are ordered by recall within each category. Overall recall intervals retain the original 10,000-resample percentile bootstrap over 16 topics.

For an exploratory comparison of category means, we first average the two cultural-arm recalls within each model and matched pair. We then average across models in each category, with equal model weights. Differences between categories are tested using all $2^8 = 256$ sign assignments over the eight pair-level differences. These two-sided tests assume independent, sign-exchangeable pair-level contrasts and hold the selected models fixed. We apply Holm correction across four overall-recall comparisons: the three pairwise category contrasts and the API group versus all eight open-weight models combined. The original benchmark-tier summaries are retained in the accompanying analysis files for traceability.

RQ3: developer-country comparisons. We compare eight models from US developers with four models from Chinese developers. Mistral, from France, is excluded from this contrast but retained in the other analyses. The US group consists of Claude Opus 4.8, GPT 5.5, Gemini 3.1 Pro, Grok 4.3, Llama 4 Maverick, GPT-OSS 120B, Gemma 4 31B, and Llama 3.3 70B. The Chinese group consists of DeepSeek V4 Pro, Qwen 3.7 Max, Kimi K2.6, and GLM 5.2.

We examine two outcomes: the difference in overall recall and the difference in WEIRD–Serbian recall gaps. For each outcome, models receive equal weight within their developer group, and the eight matched topic pairs are the blocks for exact two-sided sign-flip tests. We report 10,000-resample pair-bootstrap intervals and apply Holm correction across these two RQ3 tests. These exploratory comparisons describe the selected systems; they do not identify a causal effect of developer country.

Descriptive checks on the judging procedure. We report Fleiss' κ, Gwet's AC1, and Krippendorff's α overall and by cultural arm, together with judge-level no-match rates. Leave-one-judge-out analyses describe sensitivity to the panel's composition. These checks assess consistency and sensitivity, not whether the matching decisions are correct or free from shared bias.

4 Results

4.1 RQ1: Recall on matched WEIRD and Serbian topics

Mean recall is 0.581 for WEIRD topics and 0.594 for Serbian topics. Across the eight matched topic pairs, the mean WEIRD-minus-Serbian difference is -0.013, or -1.3 percentage points (95% bootstrap CI: -8.1 to 5.8 percentage points). The median pair-level difference is -0.014.

The exact one-sided Wilcoxon signed-rank test provides no evidence for the hypothesised WEIRD advantage ($p = 0.680$). The two-sided test gives $p = 0.742$, with a matched-pairs rank-biserial correlation of -0.17. The point estimate favours Serbian topics slightly, but the interval includes differences in both directions. This result does not establish equivalent coverage across the two settings.

Four matched pairs favour WEIRD topics and four favour Serbian topics. The largest differences point in opposite directions: recall is 19.1 percentage points higher for German immigration than for its Serbian migration-route counterpart, but 16.9 points lower for Canadian residential schools than for Srebrenica. Mean topic recall ranges from 0.449 for the Serbian migration-route topic to 0.685 for Serbia's West-versus-Russia alignment. Both extremes occur among the Serbian topics, indicating that topic-level variation does not follow a uniform cultural direction.

The additional sensitivity analyses lead to the same substantive interpretation. Averaging over pairs within each model gives one-sided $p = 0.843$ and two-sided $p = 0.331$. Independently resampling models and topic pairs gives a crossed-bootstrap interval of $[-0.090, 0.070]$. The model-based crossed-effects check estimates the same mean gap, -0.013 (SE 0.038; 95% Wald CI $[-0.088, 0.062]$; two-sided $p = 0.733$). These are supplementary checks; the eight-pair test remains the primary analysis.

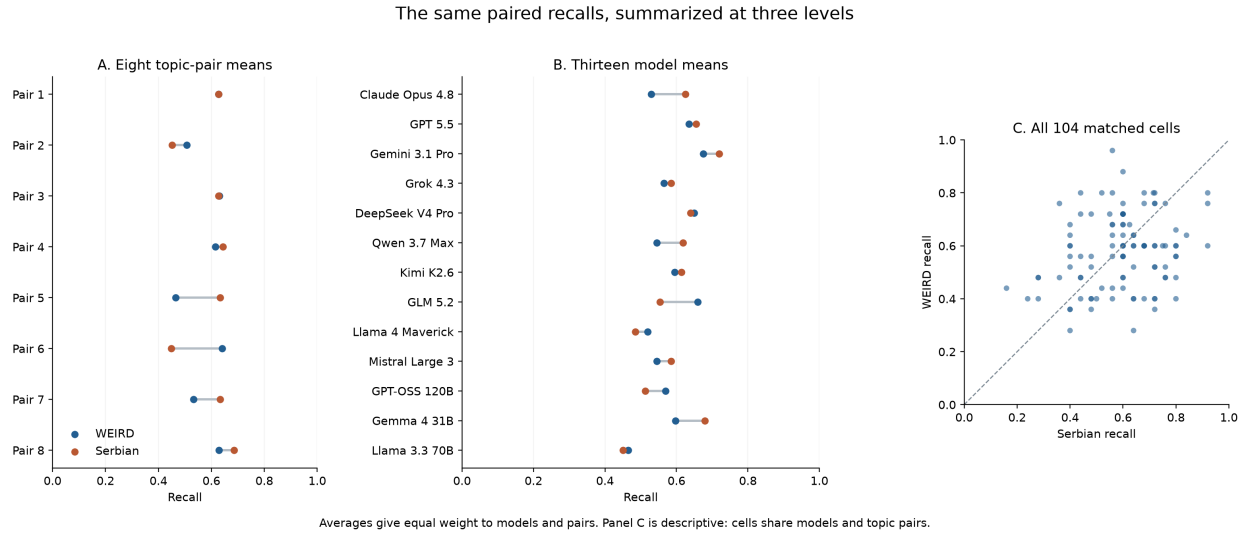


Figure 1: Paired recall at three levels. Panel A averages the 13 generators within each matched topic pair. Panel B averages the eight pairs within each generator. Panel C plots all 104 model–pair comparisons, with the dashed line indicating equal recall. Points above the line favour WEIRD topics. Panel C is descriptive: its observations share generators and matched topic pairs and are not independent replicates.

4.2 RQ2: Recall by model and access category

Table 1 reports perspective recall, with separate columns for WEIRD and Serbian topics. Overall recall ranges from 0.458 for Llama 3.3 70B to 0.698 for Gemini 3.1 Pro. Even the highest-recall model leaves approximately 30% of the reference coverage unrecovered on average.

Table 1: Perspective recall by access category, ordered by overall recall within each category. Serbian topics constitute the non-WEIRD arm. Overall intervals are the original 95% bootstrap intervals over 16 topics; cultural-arm columns are means over eight topics. The Frontier/API group includes Qwen 3.7 Max; larger and smaller open-weight systems are shown separately. These group labels are not a capability ranking.

Model	Overall recall (95% CI)	WEIRD	Serbian
Frontier/API systems (5)			
Gemini 3.1 Pro	0.698 (0.630–0.765)	0.675	0.720
GPT 5.5	0.645 (0.600–0.692)	0.635	0.655
Qwen 3.7 Max	0.582 (0.524–0.639)	0.545	0.618
Claude Opus 4.8	0.578 (0.512–0.643)	0.530	0.625
Grok 4.3	0.575 (0.522–0.625)	0.565	0.585
Open-weight systems (6)			
DeepSeek V4 Pro	0.645 (0.560–0.730)	0.650	0.640
GLM 5.2	0.607 (0.542–0.667)	0.660	0.554
Kimi K2.6	0.604 (0.548–0.656)	0.595	0.614
Mistral Large 3	0.565 (0.485–0.640)	0.545	0.585
GPT-OSS 120B	0.541 (0.468–0.612)	0.570	0.513
Llama 4 Maverick	0.502 (0.432–0.580)	0.520	0.485
Small open-weight systems (2)			
Gemma 4 31B	0.638 (0.578–0.698)	0.598	0.679
Llama 3.3 70B	0.458 (0.400–0.518)	0.465	0.450

The mean recall of the selected models is 0.615 in the Frontier/API group, 0.578 in the larger open-weight group, and 0.548 in the smaller open-weight group. Combining the eight open-weight models gives mean recall 0.570, compared with 0.615 for the five API systems, an observed difference of 4.5 percentage points. The exact pair-blocked test gives $p = 0.086$ before correction and $p = 0.258$ after Holm correction across the four RQ2 contrasts. None of those four comparisons meets the 0.05 threshold after correction; the smallest adjusted value is $p = 0.063$ for the API versus smaller open-weight comparison. These exploratory results do not establish equivalence or isolate the effects of size, openness, or capability.

Individual models also vary substantially within categories. Gemma 4 31B has recall 0.638, close to the 0.645 reported for GPT 5.5 and DeepSeek V4 Pro, whereas Llama 3.3 70B has recall 0.458. A category mean therefore does not describe every model in that category. Overlapping individual confidence intervals should not be interpreted as evidence of equivalence.

4.3 RQ3: US and Chinese developer groups

The four selected models from Chinese developers have mean overall recall 0.609, compared with 0.579 for the eight models from US developers (Table 2). The US-minus-China difference is -0.030 (95% pair-bootstrap CI $[-0.048, -0.013]$). The exploratory two-sided pair-blocked sign-flip test gives $p = 0.023$, or $p = 0.047$ after Holm correction across the two RQ3 outcomes.

Table 2: Recall by developer group. Models receive equal weight within each group. Mistral is excluded from this US–China contrast. A positive WEIRD–Serbian gap indicates higher recall on WEIRD topics.

Group	Models	Overall	WEIRD	Serbian	Gap
US	8	0.579	0.570	0.589	−0.019
China	4	0.609	0.613	0.606	+0.006

The groups' cultural gaps differ less conclusively. The mean WEIRD-minus-Serbian gap is -0.019 in the US group and $+0.006$ in the Chinese group. Their difference is -0.025 (95% pair-bootstrap CI

$[-0.082, 0.025]$; raw and Holm-adjusted $p = 0.453$). The observed overall-recall difference therefore does not establish a corresponding difference in the WEIRD–Serbian gap.

These results describe the selected systems. The Chinese group contains one API and three larger open-weight models, whereas the US group contains four API, two larger open-weight, and two smaller open-weight models. The US group also includes multiple models from some developers. Developer country is consequently entangled with model selection and category composition. The comparison does not establish a general advantage of Chinese models or a causal effect of developer location.

4.4 Descriptive checks on judge consistency

Agreement statistics have similar values for WEIRD and Serbian topics (Table 3). Fleiss' κ is 0.761 for WEIRD topics and 0.774 for Serbian topics. Gwet's AC1 and Krippendorff's α show the same descriptive pattern. The judge-level no-match rates are 32.9% and 31.1%, respectively.

Table 3: Descriptive inter-judge agreement. These statistics assess consistency of matching decisions, not correctness or absence of shared bias.

Scope	Fleiss' κ	Gwet's AC1	Krippendorff's α
All topics	0.768	0.783	0.769
WEIRD topics	0.761	0.777	0.762
Serbian topics	0.774	0.789	0.775

Removing one judge at a time leaves the highest-recall model unchanged and produces small changes in the pooled WEIRD-minus-Serbian gap, which ranges from approximately -0.020 to -0.008 across removals. These checks provide no obvious indication of substantially lower agreement for Serbian topics or dependence of the pooled result on one judge. They are descriptive checks, not formal demonstrations that the panels have equal error rates. Several judges may agree on an incorrect match, and shared cultural biases remain possible.

Conclusion: suggested updated passage

Across eight matched WEIRD–Serbian topic pairs and 13 generators, we did not detect the predicted WEIRD recall advantage. Coverage nevertheless remained incomplete in both settings, and recall varied across models and controversies. Exploratory comparisons described variation among the selected model categories and higher overall recall in the selected Chinese developer group, but did not establish a US–China difference in the cultural recall gap. The category and country comparisons are limited by the composition of the model sample. The primary methodological implication remains that perspective coverage should be measured directly using comparisons that account for topic selection.

Supplementary metrics: retain outside the main recall table

Precision describes the share of generated outputs that match any reference. In the single-judge example $(1, 2, 2, 4, \text{no match})$, precision is $4/5 = 0.80$, whereas recall is $3/5 = 0.60$. Duplicate matches can contribute to precision without adding reference coverage.

For output k, let $b_{jk} = 1$ when judge j selects any reference and $b_{jk} = 0$ for a valid no-match decision. The output's precision credit and the topic precision are

$$q_k = \frac{1}{|J_k|} \sum_{j \in J_k} b_{jk}, \qquad P_{mt} = \frac{1}{5} \sum_{k=1}^{5} q_k.$$

Model precision averages the 16 topic precisions. Model F1 is calculated from model-level mean precision and mean recall,

$$F1_m = \frac{2\overline{P}_m \overline{R}_m}{\overline{P}_m + \overline{R}_m},$$

rather than by averaging judge-level or topic-level F1 values. These metrics are supplementary because an output outside the five-reference set may still express a legitimate position. Thus, an unmatched output does not by itself establish invalid content, and the main omission analysis uses recall.